

SEIZURE

is a transient occurrence of signs and/or symptoms resulting from abnormal excessive or synchronous neuronal activity in the brain.

Simple seizure: -no alteration in consciousness.

oComplex seizure: -no alteration in consciousness.

EPILEPSY

.is present when two or more unprovoked seizures occur in a time frame of longer than 24 hours.

CAUSES OF SEIZURE

- .1. Idiopathic
- .2. CNS infections
 - . Meningitis
 - . Encephalitis
 - . Brain abscess
 - . Syphilis(tertiary)

- Metabolic

- Electrolyte imbalance (calcium, sodium, magnesium, phosphorus)

- Hypoglycemia

- 4. Structural

- Head trauma

- Brain tumor

.5. Others

- Sepsis
- Drug abuse
- Ingestion of toxins (accidental and non-accidental)

CLASSIFICATIONS OF SEIZURE

- .Seizure can be classified as
 - .o Focal: initial activation of a system of nervous limited to one part of hemisphere.
 - .o Generalized: first clinical and EEG changes indicate synchronous involvement of all both hemisphere

TYPES OF FOCAL AND GENERALIZED SEIZURES:

- Tonic: increased tone or rigidity.
- Atonic: flaccidity or lack of movement during convulsion.
- Clonic: rhythmic fast muscle contraction and slightly longer relaxation.
- Myoclonic: shock like contraction of a muscle < 50msec often repetitive.

Febrile seizure

.a seizure that occurs at 38oc or higher, that are not the result of CNS infection or any metabolic imbalance, and in the absence of prior afebrile seizures.

Clinical features

- Elevated temperature ($>38^{\circ}\text{C}$)
- Convulsions usually brief and self limiting (usually <5 minutes, and may be prolonged for more than 15 minutes in complex febrile seizure) but may recur if temperature remains high
- No neurological abnormality in the period between convulsions
- Generally benign and with good prognosis

Treatment

- .Give an antipyretic: paracetamol 15 mg/kg every 6 hours until fever subsides
- . If convulsing: Give diazepam 500 micrograms/kg rectally (using suppositories/rectal tube or diluted parenteral solution) Maximum dose is 10 mg, Repeat prn after 10 minutes

Status epilepticus

.Status epilepticus generally refers to the occurrence of a single unremitting seizure with duration longer than 30 minutes or frequent clinical seizures without an interictal return to the baseline clinical state.

treatment

.Stabilization of the airway, maintenance of adequate ventilation (with oxygen administered as necessary), and circulatory support.

- .Dextrose 10%5 mL/kg children
- . Diazepam as above in febrile convulsion,
repeated after 5-10 minIf not responsive,
Phenytoin 15-18 mg/kg over 1 hour

.If no response consider Phenobarbital 10-15 mg/kg slowly IV then 10mg per kg bid

.A longer-acting antiepileptic drug should also be administered because of the relatively short duration of action of diazepam, Phenytoin is preferred over phenobarbital

